

Risk Management and Valuation

Group Coursework 1B (Diageo)

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Executive Summary

In this report, we first identify Diageo's four key business segments - alcohol, beer, packaged food, and fast food. We then analyse their operations via several key performance indicators. Secondly, we produce a benchmark for the current market value of Diageo's equity based on the multiples valuation method (PE ratio and EV/EBIT) to identify where Diageo ranks among its peers. Third, we analyse the company's financial policy, capital structure, and growth. Finally, we propose a new capital structure that suits Diageo better, after its transition into a pure beverage alcohol business, supported by Monte Carlo simulation findings and a qualitative assessment of risks and distress costs. Our findings indicate that while Diageo's current structure is safe, it underutilises valuable tax shields and limits strategic flexibility. We recommend a target interest coverage ratio of 4-5x, accepting a credit rating downgrade to BBB in order to capture tax benefits, enhance return on equity, and maintain sufficient debt capacity for transformative acquisitions.

1 Diageo's Business and Operational Performance

1.1 Business Overview

Diageo was formed in November 1997 through the merger of Grand Metropolitan plc and Guinness plc. Following the merger, the company became one of the world's largest consumer food and drinks groups, operating in over 140 countries with annual sales exceeding £13 billion. The company was organised into four business segments.

Spirits and wine was Diageo's largest and most strategically important division. It produced and marketed a portfolio of internationally recognised premium brands including Johnnie Walker, Smirnoff, Baileys, Tanqueray and Cuervo. The segment benefited from strong brand equity, pricing power and relatively stable demand in mature markets such as Europe and North America. It was both the biggest contributor to revenues and the fastest growing division, making it the core value driver of the company.

The beer segment, led by Guinness Brewing, produced and distributed beer globally. While smaller than spirits and wine, it represented the second-largest business. Given similarities in production processes and distribution channels, management sought to integrate beer and spirits operations to generate cost efficiencies and operational synergies.

The third division operated in the branded packaged food market, with products such as Progresso and Häagen-Dazs. Although a well-established consumer business, it operated in a competitive environment structurally different from beverage alcohol and was less aligned with Diageo's premium drinks strategy.

The fast food segment consisted primarily of Burger King, operating fast-food outlets globally. While it generated substantial revenues, it belonged to a different industry with distinct operational dynamics compared to beverage alcohol.

Overall, although Diageo initially operated as a diversified food and drinks conglomerate, the com-

pany's core strategic focus increasingly centred on premium beverage alcohol, particularly the **spirits and wine** division. This represented its principal source of competitive advantage and long-term value creation.

1.2 Operational Performance

Key operating indicators are derived from the consolidated financial data presented in Appendix 1.1. A summary of the calculated operating metrics is reported in Appendix 1.2.

The operating indicators reported above suggest that Diageo's performance strengthened over FY98 - FY00 despite limited top-line growth. Revenue declined by 1.9% in FY99 before stabilising with a modest 0.6% recovery in FY00, indicating temporary volatility rather than structural deterioration.

Profitability improved steadily: operating margins remained strong at around 16% and increased to 16.7% by FY00, while net margins rose from 7.3% to 8.2%. This margin expansion suggests effective cost control, pricing discipline, and a favourable product mix.

Operational efficiency also improved, as asset turnover increased from 0.70x to 0.74x, reflecting better utilisation of the asset base. Consequently, ROA rose from 5.1% to 6.0%, indicating strengthening operating performance driven by improved margins and asset efficiency.

The evolution of ROE over the period reflects improvements in operating margins and asset efficiency, consistent with a DuPont-style decomposition where returns are primarily driven by operating performance rather than financial leverage. Return on Equity (ROE) increased from 17.0% in FY98 to 20.5% in FY99 before moderating to 18.5% in FY00. The overall improvement in ROE over the period reflects enhanced profitability and asset utilisation, suggesting that shareholder returns were supported by stronger operating fundamentals.

1.3 Segment Profitability

Segment profitability by business line is reported in Appendix 1.3, with calculated operating margins summarised in Appendix 1.4.

The spirits and wine division generates significantly higher operating margins than Beer, highlighting the profitability of premium spirits. This confirms that the company's strategic focus on beverage alcohol, particularly premium spirits, is well justified from an operational perspective.

1.4 Overall Assessment

Although revenue growth was modest during FY98 - FY00, Diageo's operational fundamentals show clear signs of strengthening. Profit margins expanded, asset utilisation improved, and ROA increased steadily, indicating that performance improvements were driven by operating quality rather than temporary financial effects.

Importantly, segment data reinforce this conclusion. The spirits and wine division as the company's largest and most profitable segment generated operating margins of approximately 20% in FY00, significantly higher than Beer. With more than 70% of sales generated in Europe and North America, Diageo operates in mature but stable markets where brand strength and pricing power are critical.

Management's strategic decision to divest non-core businesses such as packaged food and fast food further supports a clearer focus on high-margin beverage alcohol. Given the superior profitability of the Spirits segment and the improvement in operating efficiency, this strategic refocusing appears operationally justified.

Overall, despite limited top-line expansion and weaker stock market performance, Diageo demonstrates strong margins, improving efficiency, and a high-quality premium portfolio. From a business standpoint, the company exhibits solid and increasingly focused operating performance.

2 Benchmark for Current Market Value of Diageo's Equity

Since the 1997 merger of Grand Metropolitan and Guinness, Diageo's stock price has not performed well against broad market indices like the FTSE 100 Index, where the stock price plummeted in July 1999 and has lagged behind by a huge margin ever since (Appendix 2.1). To revitalise the business and the stock price, Paul Walsh, the Group Chief Executive of Diageo developed a new strategy to focus solely on the beverage alcohol business by exiting the fast-food business and selling off its packaged food subsidiary.

2.1 Assumptions and Multiples Valuation Method

While Diageo's stock price has underperformed against a broad market index, it might not be an idiosyncratic factor but instead a structural issue of the industries in which Diageo operated; the industries (alcohol and beer, for example) might have undergone a downturn and thus affected Diageo systematically. Therefore, to evaluate the performance of Diageo more objectively and rigorously, we compare Diageo to a benchmark for the current market value of Diageo's equity (as of September 2000) using the multiples valuation method.

Since Diageo's two main businesses are in alcoholic beverages and is concentrating on the beverage alcohol business, we compare Diageo to only companies in the alcohol (Allied Domecq and Pernod Ricard) and beer (Anheuser Busch, Carlsberg, Heineken) industries, excluding Seagram since its financial ratios are outliers compared to the rest of its peers to be more representative of its current and future performance. To compute the current market value of Diageo's equity, or its continuation value, we use both the Price/Earnings ratio (PE ratio) and Enterprise Value/EBIT (EV/EBIT) which are common valuation multiples (Appendix 2.2). While the multiples valuation is straightforward to calculate Diageo's implied equity value as we can value Diageo based on what the market is currently paying for similar companies, it has its limits. The multiples approach for the continuation value assumes that we expect the industry multiples to remain stable.

2.2 Analysis

Diageo's actual market value of equity for FY00 is £20.144 Billion. Compared to the implied market value of equity (£17.8 Billion) derived from the peer average PE ratio of 18.24 x, Diageo seems overvalued. However, when compared to the implied equity value (£23.4 Billion) derived from the peer average EV/EBIT ratio of 14.76 x, Diageo seems undervalued (Appendix 2.3). The implied equity value calculated from these two ratios differ as the PE ratio is sensitive to the capital structure. PE ratio uses net income, and net income is calculated after interest expense where the interest expense depends on the amount of debt a company has. So, as leverage changes, net income changes; this can distort the valuation of Diageo based on its financing decisions alone. On the other hand, EV/EBIT is independent of leverage and focuses on operating performance only. Therefore, the market value derived from EV/EBIT is the more reliable indicator.

Since the actual current market value of Diageo is below its peers from the EV/EBIT ratio, the market is conservative about Diageo's current growth and they feel that Diageo is a riskier investment compared to its peers. However, if Diageo manages to successfully restructure by consolidating its business and focus into its beverage alcohol business, there is plenty of upside of catching up to its peers in the alcohol and beer industry.

3 Financial Policy, Capital Structure, and Growth Analysis

3.1 Financial Policy from 1998 to 2000

Between 1998 and 2000, Diageo followed an actively managed, rating-targeted financial policy. Rather than targeting a specific debt-to-equity ratio, management focused on maintaining an interest coverage ratio (EBITDA/Interest) within a range of 5 to 8 times to preserve its A+ credit rating. The firm adjusted leverage in a "lumpy" manner through debt issuance and share repurchases when coverage deviated from this target band. In particular, Diageo re-levered its balance sheet in 1998 and 1999 through large share repurchase programs, distributing billions of pounds to shareholders and reducing outstanding equity by approximately 5% of issued capital.

Another important feature of Diageo's financial policy was its reliance on short-term commercial paper financing. Approximately 47% of total debt consisted of short-term commercial paper, reflecting a deliberate strategy to minimise funding costs. However, access to this market depended critically on maintaining an investment-grade credit rating. Overall, Diageo's financial policy during this period can be characterised as actively managed, with an emphasis on preserving credit quality, lowering funding costs, and retaining financial flexibility for potential acquisitions.

3.2 Interpretation of Diageo's Capital Structure

Diageo's current capital structure can be described as the outcome of a conservative and rating-constrained financial policy rather than a passive financing outcome. The firm targeted an interest coverage ratio between 5 and 8 times to preserve its A+ credit rating, and actively adjusted leverage through share repurchases and debt issuance when coverage deviated from this range.

As shown in Appendix 3.2, Diageo's interest coverage stood at 5.45× in FY2000, close to the lower bound of the target range. Market debt-to-capital was approximately 25.5%, while EBITDA-to-debt remained strong, supporting its A+ rating. Relative to peers in the alcohol and beer industry, Diageo's interest coverage ratio of approximately 5.0× was relatively conservative. Compared with peers in the alcohol and beer industry, whose coverage ratios often exceeded 10×, Diageo's coverage placed it toward the lower end of the distribution (see Appendix 3.3). The firm's ability to sustain an A+ rating at this coverage level partly reflected the stability of its branded beverage portfolio. Nevertheless, maintaining coverage near the rating boundary limited the extent to which Diageo could increase leverage to capture additional tax shields.

Throughout financial years 98, 99, and 00, operating cash flows consistently covered interest, taxes, and maintenance capital expenditures, indicating strong operating liquidity and low solvency risk (see Appendix 3.4). However, short-term debt exceeds liquidity in all three years, suggesting reliance on commercial paper roll-over and refinancing, which creates refinancing risk should the credit rating weaken. This reliance on short-term funding further reinforces the importance of maintaining an investment-grade rating. Approximately 47% of Diageo's debt was issued in the form of commercial paper, which benefited from lower borrowing costs relative to longer-term debt. But since access to this market is highly sensitive to credit quality, a downgrade from A+ to BBB or below could materially reduce market depth and increase refinancing costs, thereby constraining financial flexibility.

Taken together, Diageo's capital structure appears conservative and strongly influenced by its rating objective. Although the firm benefited from interest tax deductibility at a 27% corporate tax rate, its coverage ratio remained near the lower bound of the 5-8× target range and below several peers. This indicates that leverage was managed with reference to rating preservation and continued access to commercial paper markets, rather than solely to maximise tax shields.

3.3 Sources and Uses of Funds

Between 1998 and 2000, Diageo generated substantial operating cash flows, which constituted the primary internal source of financing. Cash flow from operations was £1,005 million in FY1998, £968 million in FY1999, and increased significantly to £1,353 million in FY2000. Relative to total assets of approximately £16-17 billion, operating cash flow represented roughly 6-8% of assets annually, which showed the firm's strong and resilient cash-generating capacity. Also, operating profits remained close to £2 billion throughout the period, indicating stable underlying profitability and solvency.

In addition to internally generated funds, Diageo also relied on debt issuance as an external source of financing, particularly short-term commercial paper. In FY1998 alone, the firm issued approximately £2,097 million in new debt. A significant portion of outstanding debt consisted of short-term commercial paper, accounting for roughly 47% of total debt, or about £3.2 billion. Asset divestitures also contributed to financing inflows, with net divestitures amounting to £1,368 million in FY1998 and remaining positive in subsequent years.

On the uses side, a significant portion of operating cash flow was allocated to shareholder distributions. Dividends were consistently high, with annual payouts around £700 million. In addition, Diageo undertook substantial share repurchase programs in 1998 and 1999 as part of its re-leveraging strategy.

Overall, Diageo's financing flows indicate that internal cash generation exceeded reinvestment needs during this period. Rather than accumulating excess cash, management chose to return funds to shareholders and adjust leverage to remain within the targeted interest coverage range. This pattern is consistent with an actively managed capital structure policy designed to balance shareholder distributions, rating preservation, and strategic flexibility.

3.4 Estimation and Comparison of the Sustainable Growth Rate

The sustainable growth rate (SGR) measures the maximum rate at which a firm can grow using internally generated funds, without issuing new equity.

For Diageo, return on equity was approximately 22.3%, and the payout ratio was around 70%. This implies a retention ratio of 30%. Therefore, the sustainable growth rate is approximately 6.7%.

To compare this with actual asset growth, total assets declined from £17,254 million in FY1998 to £16,136 million in FY2000. The cumulative asset growth over the period is approximately -6.5%.

The fact that the sustainable growth rate exceeds actual asset growth suggests that Diageo generated more internal funds than its reinvestment needs. This surplus cash flow explains the firm's share repurchases and active re-leveraging strategy during this period and also shows that Diageo's re-leveraging behaviour is deliberate rather than a passive outcome.

See Appendix 3.5 for calculations.

4 New Capital Structure Proposal

Diageo's capital structure is currently on the conservative side to maintain its strong credit rating of A+, preserve financial flexibility, support transformation, and balance tax benefits against risks, as discussed in Section 3.2. Relative to peers and given its stable cash flows, Diageo appears conservatively financed. Overall, while this approach provides rating stability, it underutilises valuable tax shields. For the focused beverage company Diageo is becoming, this structure is no longer suitable. The Monte Carlo analysis (Appendix 4.1) suggests that total costs are minimised at a coverage ratio of around 4x, implying scope for moderately higher leverage.

4.1 Business Transition

Diageo's strategy to focus exclusively on the beverage alcohol business, pursuing growth through both organic investment and acquisitions, is expected to enhance the stability and predictability of its cash flows. Organic growth initiatives, including product extensions such as Smirnoff Ice and modernisation of production facilities will require £400 to £500 million in annual capital expenditure over the next five years. At the same time, Diageo is preparing for transformative acquisitions, with potential spending of \$6 to \$8 billion over the next three years, including a joint bid for Seagrams with Pernod Ricard.

To execute this plan, Diageo requires the financial power to act decisively when opportunities arise. However, the current capital structure, while safe, is no longer sufficient to provide this level of strategic flexibility. Its conservative leverage leaves substantial tax shields untapped and limits the company's capacity to compete efficiently in a consolidating industry.

4.2 New Capital Structure

We propose a new capital structure to support Diageo's business transition by lowering its interest coverage ratio target to 4 - 5x, with an optimal point near 4x as suggested by the Monte Carlo simulation (see Appendix 4.1). The simulation reveals a U-shaped trade-off between tax payable and expected distress costs. At higher coverage levels, tax benefits are under-utilised, while at lower coverage levels expected distress costs increase non-linearly. Total expected costs are minimised at around 4x. We therefore recommend that Diageo adopt this more leveraged structure.

An interest coverage ratio (EBIT/Interest Expense) measures a company's ability to meet its interest obligations. A higher ratio indicates stronger financial health. Diageo's current coverage ratio stands at 5.4x, supporting its A+ credit rating. By comparison, some industry peers maintain coverage above 10x. Reducing Diageo's coverage to 4 - 5x would likely result in a downgrade to BBB, which is still investment grade and above average for companies with similar coverage levels.

Assuming EBIT remains unchanged in the short term, reducing the coverage ratio around 4 - 5x would require increasing outstanding debts by approximately £2 billion (see Appendix 4.3). We recommend funding this through long-term bonds rather than short-term commercial paper. This would reduce Diageo's reliance on roll-over funding and mitigate the refinancing risk identified in Section 3.2.

The increase in interest expense would allow Diageo to capture greater tax shield benefits. At a 27% tax rate, an additional debt of £2 billion could generate annual tax savings of approximately £40 million (see Appendix 4.3). Lower taxes and higher net income from growth would also improve return on equity (ROE), signalling greater efficiency.

Note that a downgrade to BBB would reduce Diageo's debt capacity from £8 billion to £5 - £8 billion. This is still sufficient to fund the acquisition and capital expenditure plans outlined for the business transition.

Based on the forecasted borrowing for FY 01 (Appendix 4.4), the increase in debt could reduce coverage to around 3.14x, below the proposed target. However, EBIT will not remain the same in practice. It is expected to grow through increased revenues from organic growth and acquisitions, operational efficiency from integration, and cost reductions from modernised production facilities. These factors will push the coverage ratio higher than the static calculation suggests. Nevertheless, Diageo should phase its borrowing carefully, structuring new debt with longer maturities to maintain coverage of 4x, particularly after the expected downgrade to BBB (likely in FY02), when the interest costs on future borrowing will rise.

4.3 Sources of Risk

Under the proposed capital structure, financial and refinancing risks are expected to increase modestly due to higher interest payments and a larger debt burden. However these risks remain manageable for two reasons. First, the tax shield from additional debt partially offsets the higher interest costs. Second, by issuing long-term bonds rather than short-term commercial paper, Diageo can reduce its exposure to roll-over risk and lock in fixed-rate funding. While a downgrade to BBB would increase borrowing costs, the strategic benefits of acquisition capacity and tax savings outweighs these elevated risks. Moreover, shifting toward long-term bonds reduces maturity concentration risk and stabilises the debt profile.

4.4 Cost of Financial Distress

In the Monte Carlo model, distress is defined as $EBIT/Interest < 1$, triggering a permanent 20% reduction in firm value. At the proposed 4-5x coverage, the probability of EBIT falling below interest remains low, making distress costs more manageable at modestly higher leverage. The simulation output (see Appendix 4.5) confirms that the optimal coverage ratio that minimises total costs (taxes + expected distress) lies between 3.6x and 4.5x.

The distress costs are classified as direct or indirect. Direct costs, such as legal and advisory fees, are generally small relative to firm value. Indirect costs are more significant and include customer behaviour, where trust in long-term products may weaken; competitor behaviour, such as price wars or aggressive marketing; and managerial distraction, which can lead to missed investment opportunities. For Diageo, however, these indirect costs are unlikely to materialise. Its brand strength and reputation provide a buffer against customer attrition and competitive aggression, which a focused management team now dedicated solely to beverage alcohol reduces the risk of distraction. Thus, even at higher leverage, expected distress costs remain low.

4.5 Final Recommendation

While a proposed coverage ratio of 4x and a corresponding downgrade to BBB would reduce Diageo's safety margin, lower its debt capacity, and increase the probability of financial distress, these drawbacks are outweighed by significant benefits. The new structure improves tax efficiency through greater interest tax shields, enhances return on equity (ROE), and supports cost efficiency and growth, making it more suitable for Diageo's focused beverage alcohol strategy and its acquisition plans. By contrast, the current conservative structure, while safe, leaves these benefits untapped and limits the company's ability to compete in a consolidating industry.

Nevertheless, Diageo should remain mindful of execution risks, particularly around acquisition integration and potential interest rate increases, to ensure a smooth transition. These risks, however, are manageable and do not diminish the net benefits of the proposed structure.

Appendix

Appendix 1.1: Key Performance Indicators

	FY98	FY99	FY00
Sales Growth	–	-1.9%	+0.6%
Operating Margin	16.1%	16.1%	16.7%
Net Margin	7.3%	8.0%	8.2%
Asset Turnover (x)	0.70	0.72	0.74
ROA	5.1%	5.8%	6.0%
ROE	17.0%	20.5%	18.5%

Appendix 1.2: Grand Metropolitan, Guinness, and Diageo's Financial Statements (£ millions)

	GrandMet CY 97 PF	Guinness CY 97 PF	FY 97 PF	FY 98	FY 99	FY 00
Sales	8,045	4,539	12,985	12,029	11,795	11,870
Operating costs	6,978	3,584	10,982	10,659	10,278	10,088
Interest payable (net)	153	81	268	360	324	363
Associates (net)	5	71	89	112	95	121
Other	(229)	(320)	(618)	478	94	(163)
Taxes	249	255	532	721	440	401
Net Income	441	370	674	879	942	976
Dividends			654	835	674	713
	See Note			FY98	FY99	FY00
Operating profits				1,866	1,966	2,043
Cash interest paid (net)				(258)	(432)	(405)
Taxes paid				(603)	(566)	(285)
Cash From Operations				1,005	968	1,353
Dividends from investments				120	58	64
Net capital sales (expenditures)				(370)	(444)	(488)
Net divestitures (acquisitions)				1,368	121	487
Cash From Investments				1,118	(265)	63
Management of liquid resources ^a				(600)	2,195	(219)
Dividends paid				(695)	(702)	(710)
Paid in capital from shares issued (repurchased)				(2,662)	(1,336)	(80)
New debt				2,097	(716)	(544)
Cash From Financing				(1,860)	(559)	(1,553)
Total change in cash				263	144	(137)
	GrandMet CY 97 PF	Guinness CY 97 PF	FY 97 PF	FY 98	FY 99	FY 00
Assets						
Cash and marketable securities	1,786	454	2,173	2,987	1,097	1,063
Accounts receivable	2,243	1,432	3,216	3,054	3,259	3,071
Inventory	995	1,909	2,374	2,236	2,202	2,139
Fixed assets	5,513	4,105	9,625	8,977	9,720	9,863
Total Assets	10,537	7,900	17,388	17,254	16,278	16,136
Liabilities and Owners Equity						
Accounts payable	1,833	1,783	2,930	3,524	3,532	3,275
Short term debt	1,730	1,289	2,293	4,724	3,905	3,066
Long term debt	2,515	760	4,190	3,137	3,495	3,816
Other	463	167	674	705	753	694
Total Liabilities	6,541	3,999	10,087	12,090	11,685	10,851
Minority interests	416	131	530	535	567	574
Common stock	531	476	1,001	1,139	992	990
Retained earnings	3,049	3,294	5,770	3,490	3,034	3,721
Total Equity	3,996	3,901	7,301	5,164	4,593	5,285
Total Liabilities and Owners Equity	10,537	7,900	17,388	17,254	16,278	16,136
Shares Outstanding	2,119 ^b	1,877 ^b		3,880	3,402	3,397
Market Value of Equity	12,565 ^b	11,209 ^b		27,548	22,538	20,144

Appendix 1.3: Segment Operating Margins (FY00)

Segment Operating Margin (FY00)	
Spirits & Wine	20.1%
Beer	13.2%

Appendix 1.4: Diageo segment breakdown (£ millions)

Turnover	CY 96 PF	FY 97 PF	FY 98	FY 99	FY 00
Class of business					
Spirits and Wine	5,830	5,692	5,327	4,929	4,971
Beer	2,262	2,259	2,176	2,234	2,146
Packaged Food	3,784	3,755	3,654	3,757	3,812
Restaurants	877	879	869	875	941
Associates and other ^a	687	400	3	0	0
Total	13,440	12,985	12,029	11,795	11,870
Geographical Area					
Europe	4,556	4,443	4,262	4,230	4,181
North America	5,790	5,718	5,619	5,656	5,639
Asia Pacific	1,260	1,226	915	777	886
Rest of World	1,147	1,198	1,230	1,132	1,164
Associates and other ^a	687	400	3	0	0
Total	13,440	12,985	12,029	11,795	11,870
Operating Profit	CY 96 PF	FY 97 PF	FY 98	FY 99	FY 00
Class of business					
Spirits and Wine	1,138	1,135	1,070	967	1,002
Beer	254	264	247	273	284
Packaged Food	412	423	447	478	492
Restaurants	161	160	179	185	202
Associates and other ^a	36	21	(1)	0	0
Total	2,001	2,003	1,942	1,903	1,980
Geographical Area					
Europe	543	552	534	594	585
North America	913	901	938	936	956
Asia Pacific	251	261	174	131	170
Rest of World	258	268	297	242	269
Associates and other ^a	36	21	(1)	0	0
Total	2,001	2,003	1,942	1,903	1,980

Appendix 1.5: Detailed Calculations of Operating Indicators (FY98–FY00)

Sales Growth

$$\text{Sales growth}_{99} = \frac{11,795 - 12,029}{12,029} = -0.0194 \approx -1.9\%$$

$$\text{Sales growth}_{00} = \frac{11,870 - 11,795}{11,795} = 0.00636 \approx +0.6\%$$

Operating Margin

$$\text{Op. margin}_{98} = \frac{1,942}{12,029} = 16.1\%$$

$$\text{Op. margin}_{99} = \frac{1,903}{11,795} = 16.1\%$$

$$\text{Op. margin}_{00} = \frac{1,980}{11,870} = 16.7\%$$

Net Margin

$$\text{Net margin}_{98} = \frac{879}{12,029} = 7.3\%$$

$$\text{Net margin}_{99} = \frac{942}{11,795} = 8.0\%$$

$$\text{Net margin}_{00} = \frac{976}{11,870} = 8.2\%$$

Asset Turnover

$$\text{AT}_{98} = \frac{12,029}{17,254} = 0.70$$

$$\text{AT}_{99} = \frac{11,795}{16,278} = 0.72$$

$$\text{AT}_{00} = \frac{11,870}{16,136} = 0.74$$

Return on Assets (ROA)

$$\text{ROA}_{98} = \frac{879}{17,254} = 5.1\%$$

$$\text{ROA}_{99} = \frac{942}{16,278} = 5.8\%$$

$$ROA_{00} = \frac{976}{16,136} = 6.0\%$$

Equity Multiplier

$$EM_{98} = \frac{17,254}{5,164} = 3.34$$

$$EM_{99} = \frac{16,278}{4,593} = 3.55$$

$$EM_{00} = \frac{16,136}{5,285} = 3.05$$

Return on Equity (ROE)

$$ROE_{98} = \frac{879}{5,164} = 17.0\%$$

$$ROE_{99} = \frac{942}{4,593} = 20.5\%$$

$$ROE_{00} = \frac{976}{5,285} = 18.5\%$$

DuPont Verification

$$ROE_{98} \approx (7.3\%) \times (0.70) \times (3.34) \approx 17.1\%$$

$$ROE_{99} \approx (8.0\%) \times (0.72) \times (3.55) \approx 20.4\%$$

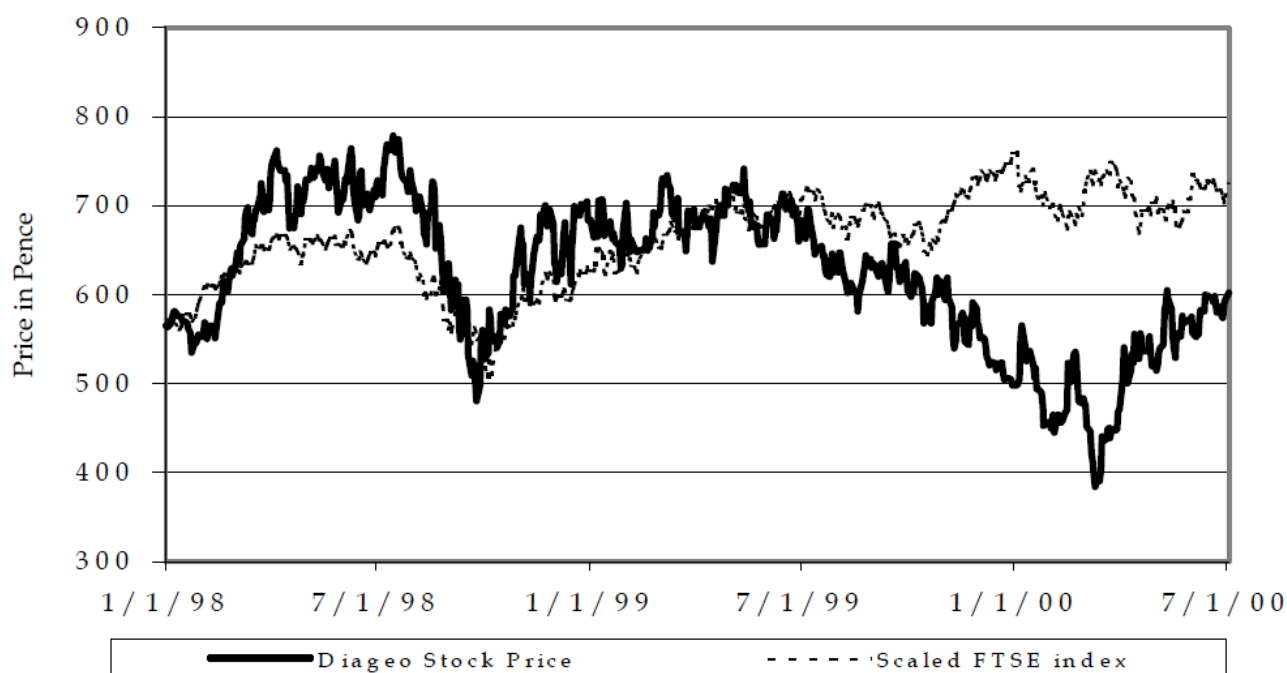
$$ROE_{00} \approx (8.2\%) \times (0.74) \times (3.05) \approx 18.5\%$$

Segment Operating Margin

$$\text{Spirits \& Wine margin} = \frac{1,002}{4,971} = 20.1\%$$

$$\text{Beer margin} = \frac{284}{2,146} = 13.2\%$$

Appendix 2.1: Diageo stock price and scaled Financial Times Stock Exchange 100 Index from January 1998 to July 2000



Appendix 2.2: Selected financial data on companies which are Diageo's rivals, as of latest annual reporting period before October 2000 (local currency)

Company	Diageo	Alcohol		Beer			Conglomerate	
		Allied Domecq	Pernod Ricard	Anheuser Busch	Carlsberg	Heineken	Seagram	Unilever
Domicile	UK	UK	France	US	Denmark	Netherlands	US	UK
Employees ('000)	72	40	14	24	18	37	34	255
Market capitalization (MM)	20,292	3,416	3,203	32,681	16,245	15,184	25,352	13,262
Market capitalization (\$ MM)	30,720	4,970	3,210	32,681	2,327	15,219	25,352	21,374
Enterprise value ^a (MM)	27,113	4,780	4,388	37,804	23,727	15,955	33,229	16,239
Enterprise value (\$ MM)	41,046	6,955	4,398	35,344	3,399	15,992	33,229	26,172
Price / Earnings ratio	20.8 x	10.8 x	13.7 x	23.3 x	14.0 x	29.4 x	1690.1 x	7.3 x
Market / Book ratio	4.3 x	18.0 x	1.6 x	8.3 x	1.6 x	5.8 x	2.1 x	2.7 x
Enterprise Value / EBIT ^b	13.7 x	11.4 x	11.8 x	16.4 x	14.2 x	20.0 x	47.9 x	5.7 x

Appendix 2.3: Benchmark for Diageo's current market value of equity via multiples valuation

Table 1: Benchmark for Diageo's current market value of equity via multiples valuation

Financial Ratio	Peer Average	Diageo	Market Value of Diageo's Equity
Price / Earnings	18.24 x	20.8 x	£17.8 Billion
Enterprise Value / EBIT	14.76 x	13.7 x	£23.4 Billion

Appendix 2.4: Calculation of multiples valuation for Diageo's benchmark

To find the peer average for the Price / Earnings (PE) ratio, we sum up the individual companies' PE ratio and divide by the number of firms. We use data from Allied Domecq, Pernod Ricard, Anheuser Busch, Carlsberg and Heineken as these are Diageo's main competitors. While Seagram is also a competitor, it is a conglomerate with a PE ratio of 1690, which is too far an outlier compared to its peers. As such, I have not chosen to include its PE ratio for calculation of Diageo's benchmark. Thus, the PE ratio (peer average) is calculated via data from Appendix 2.2 as:

$$\text{PE ratio (peer average)} = \frac{10.8 + 13.7 + 23.3 + 14.0 + 29.4}{5} = 18.24$$

The Enterprise Value / EBIT (peer average) can also be calculated the same way:

$$\text{Enterprise Value / EBIT (peer average)} = \frac{11.4 + 11.8 + 16.4 + 14.2 + 20.0}{5} = 14.76$$

Appendix 2.5: Calculation for current market value of Diageo's equity

The PE ratio is defined as $\frac{\text{market value per share}}{\text{earnings per share}}$ where earnings refer to net income. Therefore, the market value of Diageo can be calculated as below using the peer average PE ratio and net income from FY00 in Appendix 1.2:

$$\text{Market value} = \text{earnings (net income)} \times \text{PE ratio}$$

$$\text{Market value} = \text{£976 Million} \times 18.24 = \text{£17.8 Billion}$$

In order to calculate the market value of Diageo's equity using the EV/EBIT ratio, we invoke the core formula:

$$\text{Equity Value} = \text{Enterprise Value} - \text{Total Debt} + \text{Cash} \pm \text{Adjustments}$$

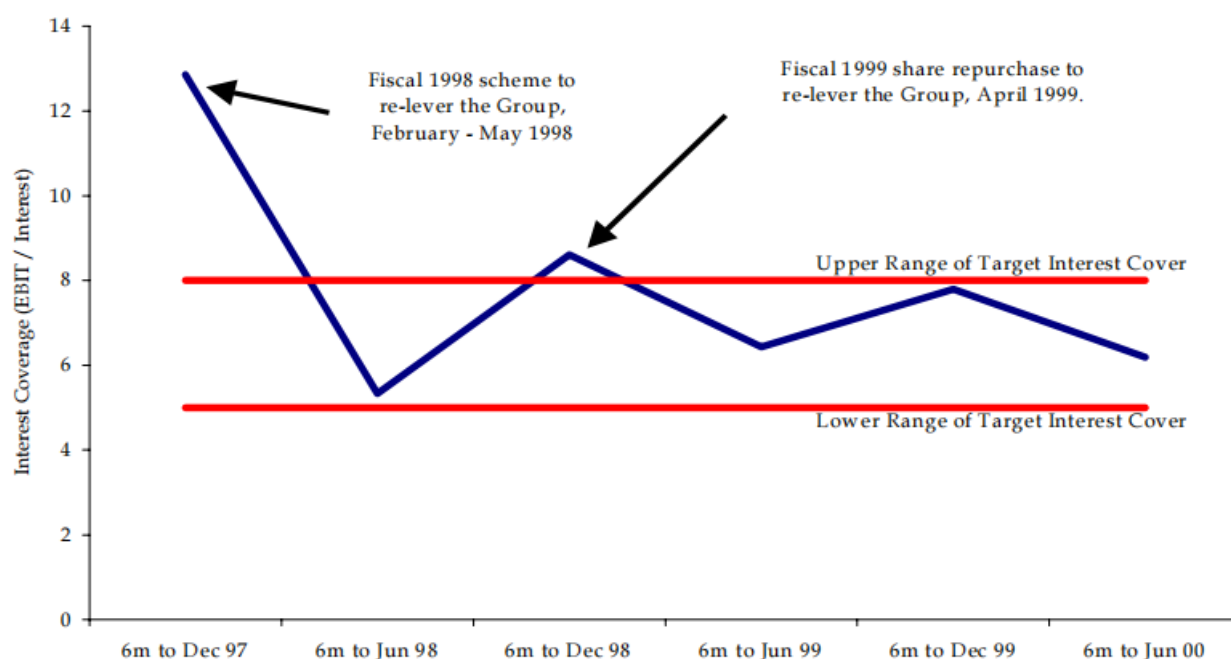
In our case, EBIT = operating profit for FY00, which can be found in Appendix 1.4. Total debt is the sum of the short term debt + long term debt in the balance sheet. Cash is under assets in the same balance sheet and is labelled as "cash and marketable securities".

$$\text{Enterprise Value} = \text{EV/EBIT} \times \text{Operating Profit}$$

$$\text{Enterprise Value} = 14.76 \times \text{£1980 Million}$$

$$\text{Equity Value} = 14.76 \times \text{£1980 M} \times (-\text{£3066M} - \text{£3816M}) + \text{£1063M} = + \text{£23.4B}$$

Appendix 3.1: Diageo interest coverage, 1997-2000



Appendix 3.2: Leverage and Financing Risk (FY 2000)

Metric (£m)	FY 00
Interest Coverage	5.45x
Market debt-to-capital	25.5%
Credit Rating	A+
Tax Shield	27%
EBITDA / Total Debt	34%
WACC	7.38% (assuming market risk premium of 5%)

Appendix 3.3: Interest Coverage Ratios in the Alcohol and Beer Industry

Company	Interest Coverage (x)	Credit Rating
Diageo	5.0	A+
Allied Domecq	5.7	A-
Pernod Ricard	10.4	NR
Anheuser-Busch	10.6	A+
Carlsberg	4.1	BBB
Heineken	15.3	AAA

Appendix 3.4: Liquidity Risk Indicators (£m)

Metric (£m)	FY 98	FY 99	FY 00
Short-term debt	4724	3095	3066
Cash and marketable securities	2987	1097	1063
Cash flow from operations	1005	968	1353
Cash and marketable securities + CFO	3992	2065	2416
Interest	360	324	363
Taxes	721	440	401
Capital expenditures	370	444	488

Appendix 3.5: Calculation of SGR

$$SGR = ROE \times (1 - \text{Payout Ratio}) \quad (1)$$

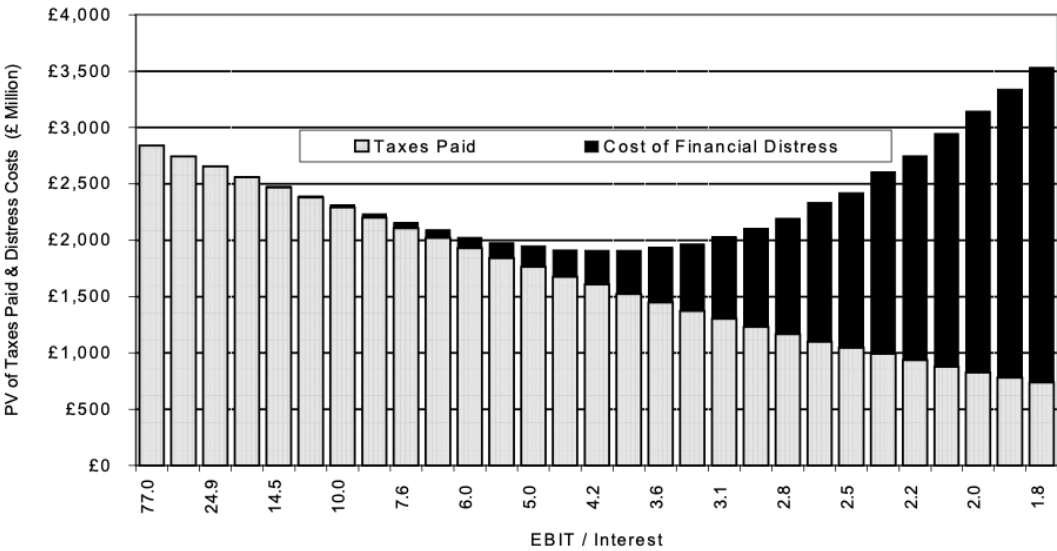
where ROE is the return on equity and the payout ratio represents the proportion of earnings distributed as dividends.

$$SGR = 0.223 \times 0.30 = 0.0669 \approx 6.7\% \quad (2)$$

$$\text{Asset Growth Rate} = \frac{16,136 - 17,254}{17,254} \approx -6.5\% \quad (3)$$

Appendix 4.1: Monte Carlo Output

Figure 2 Output chart from model.



Source: Diageo.

Note: Horizontal axis is decreasing in interest coverage, or increasing in gearing. The vertical axis is the present value of expected (average) taxes paid by Diageo plus the present value of the expected(average) cost of financial distress from the model.

The output chart shows the U-shaped total cost, with the minumum lying around 4x coverage, as argued.

Appendix 4.2: Rates for Differently Rated Companies

Exhibit 5 Selected data on bond market for 5-year notes, October 31, 2000.

Rating	Interest rate (Pounds)	Interest rate (Dollars)	Median interest coverage ^a	Issuing firms with rating	Average monthly issuance (\$ Billion)	Minimum monthly issuance (\$ Billion)	Maximum monthly issuance (\$ Billion)
UK Government	5.56%	-	-	-	-	-	-
US Government	-	5.83%	-	-	-	-	-
AAA	6.13%	6.76%	30.07	4%	3.64	0.53	11.08
AA	6.37%	6.81%	11.37	13%	3.25	0.00	23.63
A	6.76%	7.15%	8.34	25%	6.47	0.69	30.65
BBB	7.16%	7.57%	4.94	22%	1.76	0.09	8.73

Sources: Bloomberg, Securities Data Corporation, Standard and Poor's "Ratings Performance 1999: Stability and Transition."

Note: Monthly issuance data collected between January 1, 1998, and October 31, 2000, for dollar denominated bonds in the US. The global market for dollar denominated bonds was about half the size of the US market. Issuers, however, tended to be higher rated firms or governments, and the individual issue sizes larger, relative to the US market.

^a Interest coverage defined as EBITDA/interest expense.

The key rates required for the analysis are the £ and \$ rate for A and BBB rated companies. Note that Diageo can borrow at A rates before any downgrades.

Appendix 4.3: Debt Required and Tax Savings

Information: Proposed interest coverage ratio = 4x EBIT = Operating Profit = £2,043 million (see Appendix q2 4)

$$\text{Interest Coverage Ratio} = \frac{\text{EBIT}}{\text{Interest Expense}}$$

$$4 = \frac{2043}{\text{Interest Expense}}$$

⇒ Interest expense to meet the proposed interest coverage ratio is £510.75 million

The existing interest payable in FY00 is £363 million (see Appendix 1.2), hence the new interest payable is £147.75. With the interest rate for A+ rated companies at 6.76% (see Appendix 4.2), Diageo would have to borrow an additional of £2,185.65 million ≈ £2 billion.

The new interest expense gives Diageo tax saving of:

$$\begin{aligned}
 \text{Tax Savings} &= \text{Change in Interest Expense} \times \text{Marginal tax rate} \\
 &= 147.75 \text{ million} \times 27\% \\
 &= 39.89 \text{ million} \\
 &\approx 40 \text{ million}
 \end{aligned}$$

Appendix 4.4: Borrowing Forecast and Interest Coverage Ratio

The expected borrowing activity for Diageo is as such:

Activity	Description	Borrowing Amount (£)
Advertising Budget	-	1bn
Organic Growth	£400 - 500m p.a. for five years	450m
Acquisition	\$6 - 8bn in the next three years inc. joint bid of Seagram	\$4bn $\approx$ 2.64 bn

Remarks: We estimate the amount of borrowing for organic growth and acquisition (joint bid of Seagram) using the median value of the expected range.

The new interest payable after the borrowing, at the 6.76% for borrowings in pounds and at 7.15% for borrowings in dollars (Appendix 4.2) equals to:

$$\begin{aligned}
 \text{New interest payable} &= (1000 + 450) \text{ m} \times 6.76\% + \$4 \text{ bn} \times 7.15\% \\
 &\approx 286.92 \text{ m} \\
 \text{Total interest payable} &\approx 286.92 \text{ m} + 363 \text{ m} \\
 &= 649.92 \text{ m} \\
 \text{Interest coverage ratio} &= \frac{2,043 \text{ m}}{649.92 \text{ m}} \\
 &\approx 3.14x
 \end{aligned}$$

Appendix 4.5: Monte Carlo Output Analysis

Looking at the output chart in Appendix 4.2, we constructed a table to identify the coverage ratio that minimises Diageo's total cost:

EBIT / Interest	Taxes Paid (£m)	Cost of Financial Distress (£m)	Total Cost
8.8	2,200	2,250	4,450
7.6	2,100	2,150	4,250
6.8	2,000	2,100	4,100
6.0	1,900	2,010	3,910
5.5	1,800	1,990	3,790
5.0	1,750	1,950	3,700
4.6	1,690	1,900	3,590
4.2	1,600	1,900	3,500
3.9	1,500	1,900	3,400
3.6	1,475	1,950	3,425
3.3	1,450	1,990	3,400
3.1	1,350	2,100	3,450

We see that the total cost is minimised when the interest coverage ratio is at $3.9x \approx 4x$.